

- RAG Learning

6.1 Sampling, Charts & Scatter Graphs

Pearson Edexcel GCSE Mathematics 1MA1 — Statistics — spec refs S1, S2, S5, S6 — Papers 1 (non-calculator), 2 and 3 (calculator)

In 1MA1, statements S1, S2, S5 and S6 appear word for word in both the Foundation and the Higher content lists, so **everything on this page is assessed on both tiers** — there is no Higher-only material here. The topic covers collecting data from samples, using a sample to describe a population, displaying data in tables and charts, and analysing pairs of variables with scatter graphs.

6.1.1 Populations and Samples (S1)

Population — every person or item you want information about (e.g. all the customers of a shop, every egg laid on a farm this week).

Sample — the part of the population that is actually asked, measured or tested.

Census — a survey that includes every member of the population.

Why take a sample?

- It saves **time** and **money**.
- The population may be very large, or it may not be possible to contact everyone.
- Some tests **destroy** the item (e.g. checking how long a battery lasts, or the breaking strength of a rope).

Limitations of sampling

- A sample only gives an **estimate**; a second sample would probably give a slightly different answer.

- A **small** sample is less likely to represent the population.
- A **biased** sample over- or under-represents some groups, so conclusions may be wrong however large it is.

Random sample — every member of the population has the **same chance** of being chosen.

Method: make a list of the whole population (the *sampling frame*) and number it, e.g. 1 to 1200. Use a calculator's random number key or a computer to generate numbers, and select those members, ignoring any repeats, until you have the sample size you need.

Biased sample — a sample that does not fairly represent the population.

Spotting bias

How the sample was chosen	Why it is biased
Asking people in a library how often they read	Library users probably read more than average
Asking the first 15 people to arrive at a gym on Monday	Only one day and time; early Monday users may be different from others
A phone-in or website poll	Only people who choose to respond are included, often those with strong views
Asking only your own class or friends	One age group; not chosen at random
Checking only the last batch made each day	Machines or workers may perform differently at other times

Exam Tip: "It is biased" earns nothing on its own. Explain *who* is over- or under-represented, in the context of the question, then suggest an improvement: **choose randomly** from a list of the whole population, **use a larger sample**, or **include all groups/times/places**.

6.1.2 Using a Sample to Describe a Population (S1, S5)

If a sample is random and reasonably large, the **proportion** with a property in the sample is used as an estimate of the proportion in the whole population.

$$\text{Estimate for the population} = \frac{\text{number in the sample with the property}}{\text{sample size}} \times \text{population size}$$

Worked Example: A school has 1350 pupils. In a random sample of 60 pupils, 16 buy a school lunch every day. Work out an estimate for the number of pupils in the school who buy a school lunch every day.

$$\frac{16}{60} \times 1350 = \frac{21\,600}{60} = 360 \text{ pupils.}$$

Assumption: the sample is representative of all pupils in the school.

Worked Example: A farm packs 7200 eggs a week. A random sample of 150 eggs contains 4 cracked eggs. Estimate the number of cracked eggs packed in a week.

$$\frac{4}{150} \times 7200 = 4 \times 48 = 192 \text{ cracked eggs.}$$

With only 4 cracked eggs in the sample, the estimate could easily be out by quite a lot — a larger sample would make it more reliable.

6.1.3 Frequency Tables and Two-Way Tables (S2)

Categorical data — non-numerical categories (e.g. type of phone, eye colour).

Discrete numerical data — numbers that can only take particular values, usually counted (e.g. number of siblings).

A **frequency table** shows how often each category or value occurs. In a **tally**, every fifth mark is drawn across the previous four to make a "gate".

Worked Example: The number of siblings of 24 pupils:

Siblings	Tally	Frequency
0		5
1		9
2		6
3		3
4		1
Total		24

Check: $5 + 9 + 6 + 3 + 1 = 24$. Pupils with at least 2 siblings: $6 + 3 + 1 = 10$, which is $\frac{10}{24} = \frac{5}{12}$ of the pupils.

Worked Example: Two-way table. 50 students were asked how they travel to school.

	Bus	Walk	Total
Year 10	12	14	26
Year 11	15	9	24
Total	27	23	50

If only some cells are given, use the totals: e.g. Year 11 walkers = $24 - 15 = 9$, and check $14 + 9 = 23$.

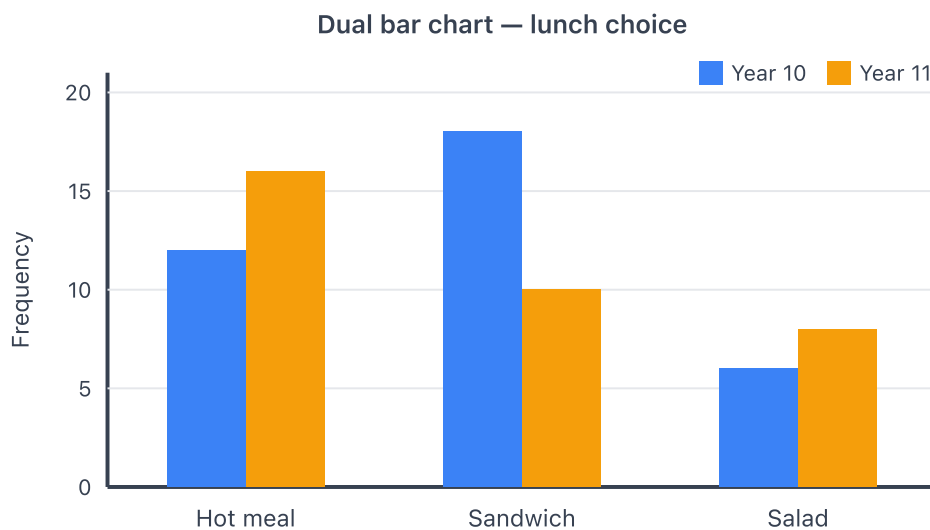
6.1.4 Bar Charts and Vertical Line Charts (S2)

Bar chart — bars of **equal width** separated by **equal gaps**; bar heights show frequencies. Used for categorical data and ungrouped discrete data.

Dual bar chart — bars for two groups side by side in each category, with a key.

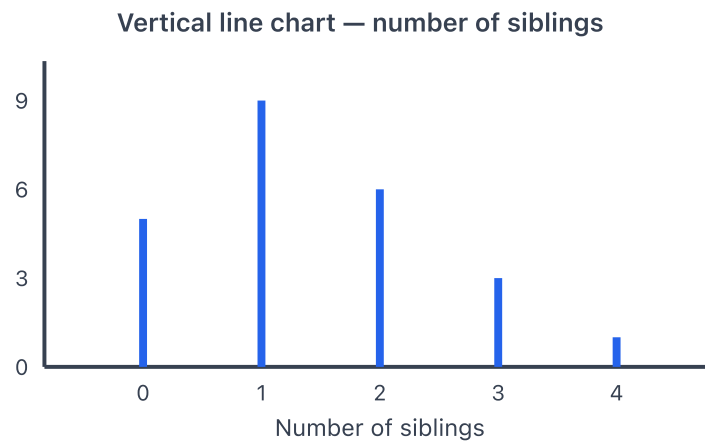
Composite bar chart — one bar per group, split into sections that add up to the group total.

Vertical line chart — thin vertical lines instead of bars; used for **ungrouped discrete numerical data**.



Worked Example: Using the dual bar chart: Year 10 — hot meal 12, sandwich 18, salad 6; Year 11 — hot meal 16, sandwich 10, salad 8.

- (a) How many students were surveyed? $(12 + 18 + 6) + (16 + 10 + 8) = 36 + 34 = 70$.
- (b) What fraction of Year 11 chose a hot meal? $\frac{16}{34} = \frac{8}{17}$.
- (c) Compare the two years: Year 10's most popular choice was a sandwich; Year 11's was a hot meal.



Exam Tip: When drawing a chart: label both axes, use an even scale that starts at 0, keep bars the same width, and leave equal gaps between bars for categories. Add a key to any dual or composite bar chart.

6.1.5 Pictograms (S2)

A **pictogram** uses a symbol to stand for a fixed number of items. It must have a **key**. A fraction of a symbol stands for the same fraction of that number.

Worked Example: Key: one cup symbol represents 8 cups of coffee sold.

- Wednesday shows $3\frac{3}{4}$ symbols: $3.75 \times 8 = 30$ cups.
- Thursday sold 20 cups: $20 \div 8 = 2.5$, so draw 2 whole cups and half a cup.
- Friday sold 14 cups: $14 \div 8 = 1.75$, so draw 1 whole cup and three-quarters of a cup.

Exam Tip: Work out what *one* symbol represents before reading anything else; if the key is missing, use a row where the frequency is given (e.g. 3 symbols = 24, so 1 symbol = 8).

6.1.6 Pie Charts (S2)

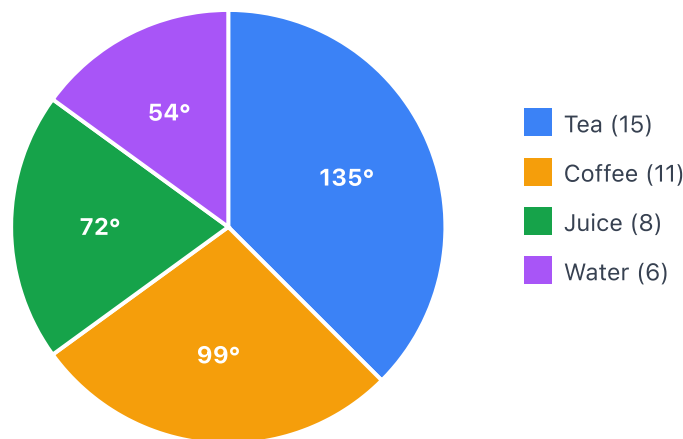
A **pie chart** shows how a total is shared between categories. Each sector's **angle** is proportional to its frequency, and the angles add up to 360° . Pie charts show *proportions*; they do not show the actual numbers unless the total is known.

$$\text{Angle} = \frac{\text{frequency}}{\text{total}} \times 360^\circ \quad \text{Frequency} = \frac{\text{angle}}{360^\circ} \times \text{total}$$

Worked Example: 40 people were asked for their favourite drink: tea 15, coffee 11, juice 8, water 6.

$$360 \div 40 = 9^\circ \text{ per person.}$$

Drink	Frequency	Angle
Tea	15	$15 \times 9 = 135^\circ$
Coffee	11	$11 \times 9 = 99^\circ$
Juice	8	$8 \times 9 = 72^\circ$
Water	6	$6 \times 9 = 54^\circ$
Total	40	360°



Worked Example: Reading a sector. A pie chart shows 200 people. The "cycle" sector is 81° .

$$\frac{81}{360} \times 200 = 45 \text{ people.}$$

Finding the total. A 54° sector represents 21 people. Then 1° represents $21 \div 54 = \frac{7}{18}$ of a person, so the total is $\frac{7}{18} \times 360 = 140$ people. (Or: $360 \div 54 = \frac{20}{3}$, and $21 \times \frac{20}{3} = 140$.)

Worked Example: Comparing pie charts. School X (300 students) has a 72° sector for "drama club". School Y (720 students) has a 40° sector.

$$\text{X: } \frac{72}{360} \times 300 = 60 \text{ students.} \quad \text{Y: } \frac{40}{360} \times 720 = 80 \text{ students.}$$

School Y has **more** students in drama club even though its sector is smaller — it is a smaller *proportion* of a much larger total.

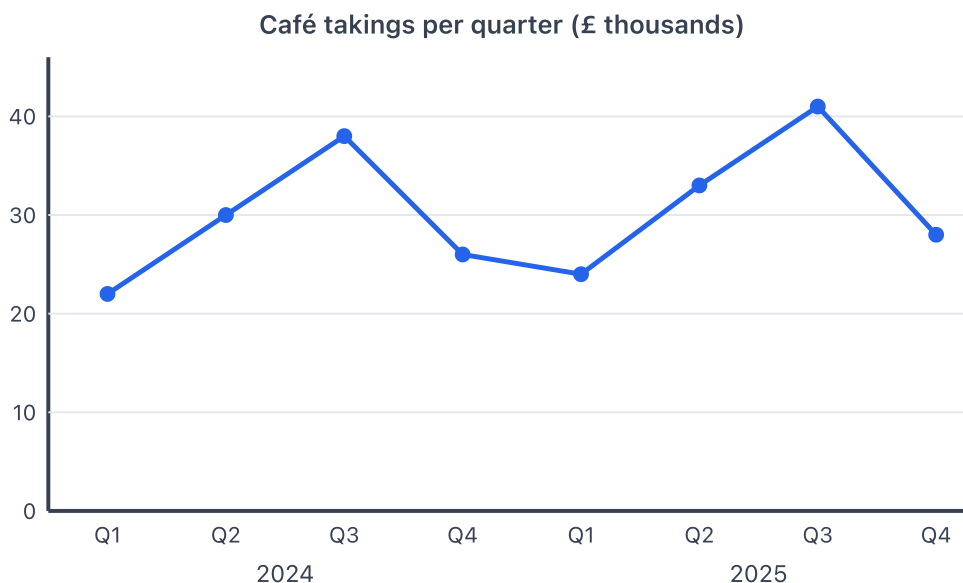
Exam Tip: On Paper 1 (non-calculator) start with "degrees per person" = $360 \div \text{total}$. Check the angles add to 360° , draw them with a protractor to within 2° , and label each sector.

6.1.7 Time Series Graphs (S2)

A **time series graph** plots data collected at regular time intervals, with time on the horizontal axis. Consecutive points are joined with straight lines.

Trend — the overall direction over time (increasing, decreasing or roughly constant).

Seasonal variation — a pattern that repeats in every cycle (every year, week, day...).



Worked Example: Takings (£ thousands): 2024 — 22, 30, 38, 26; 2025 — 24, 33, 41, 28.

Seasonal variation: takings rise to a peak in Q3 (summer) and are lowest in Q1 every year.

Trend: every quarter of 2025 is higher than the same quarter of 2024, and the yearly total rose from $22 + 30 + 38 + 26 = 116$ to $24 + 33 + 41 + 28 = 126$ thousand pounds — an increasing trend.

Percentage increase in Q3: $\frac{41 - 38}{38} \times 100 = 7.9\%$ (1 d.p.).

Exam Tip: Comparing Q3 2024 (£38k) with Q1 2025 (£24k) and saying "takings are falling" is wrong — that drop is seasonal. Judge the trend by comparing the **same season** in different years, or yearly totals.

6.1.8 Choosing and Criticising Diagrams (S2)

Data	Appropriate diagram	Example
Categorical	Bar chart, pictogram, pie chart	Favourite type of film
Two groups compared by category	Dual or composite bar chart	Lunch choice by year group
Shares of a total	Pie chart	How a household budget is spent
Ungrouped discrete numerical	Vertical line chart (or bar chart)	Number of siblings
Values over time	Time series line graph	Quarterly takings
Two numerical variables per item	Scatter graph	Hours of practice and typing speed
Grouped or continuous	Frequency polygon, grouped bar chart; histogram and cumulative frequency at Higher (see 6.2)	Journey times

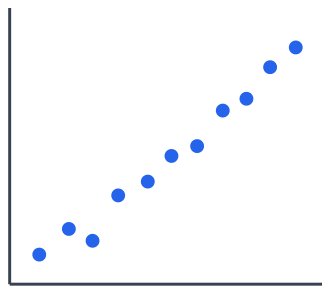
Exam Tip: You may be asked why a chart is **misleading**. Look for: a vertical axis that does not start at 0 (so small differences look huge), an uneven scale, bars or pictures of different widths, 3D effects, missing labels or keys, and pie charts with no total.

6.1.9 Scatter Graphs and Correlation (S6)

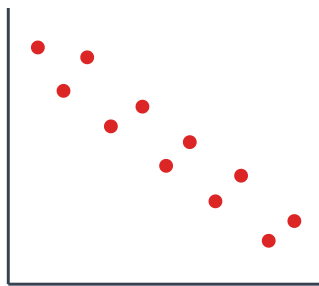
Bivariate data gives two values for each person or item. A **scatter graph** plots each pair as a point to see whether the variables are related.

- **Positive correlation** — as one variable increases, the other tends to increase.
- **Negative correlation** — as one variable increases, the other tends to decrease.
- **No correlation** — no linear relationship between the variables.

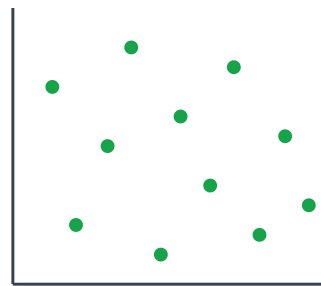
- **Strong** correlation: points close to a straight line. **Weak**: more scattered, but a trend is still visible.



Strong positive



Weak negative



No correlation

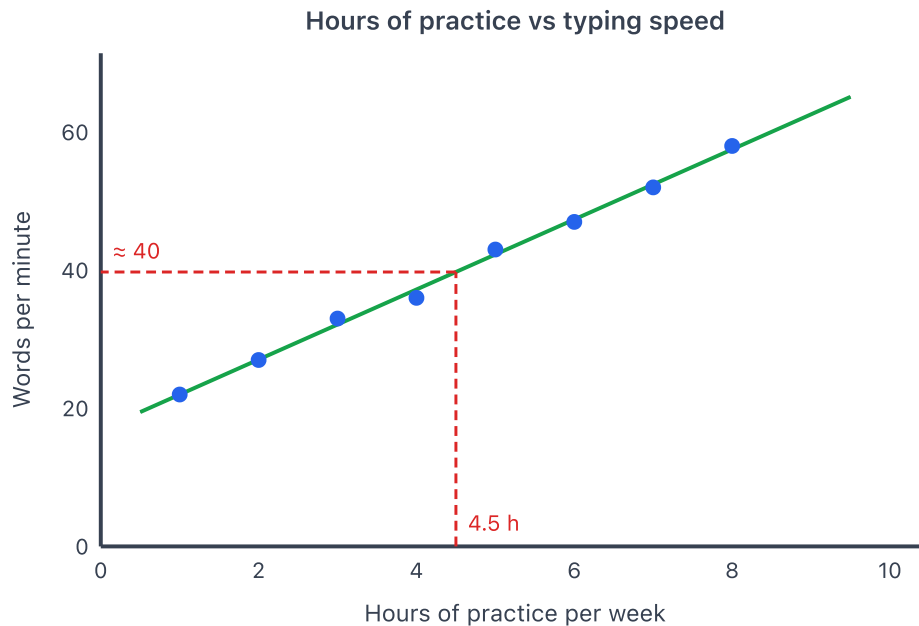
Correlation is not causation

A correlation shows that two variables are related, not that one *causes* the other. Often a **third variable** drives both.

- Umbrella sales and the number of people wearing wellies are positively correlated — rain causes both.
- Across towns, the number of cafés and the number of road accidents are positively correlated — bigger towns have more of both.

Lines of best fit

A **line of best fit** is a single straight ruled line that follows the trend, with the points roughly balanced either side of it. It need not go through the origin or through any of the plotted points. Ignore outliers when placing it, and only draw one when there is correlation.



Worked Example: Data (hours, words per minute): (1, 22), (2, 27), (3, 33), (4, 36), (5, 43), (6, 47), (7, 52), (8, 58).

Correlation: strong positive — people who practise more tend to type faster.

Estimate for 4.5 hours: go up from 4.5 to the line and across: about **40 words per minute** (a mark scheme would accept roughly 37–43 from a sensible line).

Interpolation and extrapolation

Interpolation — estimating *inside* the range of the data; fairly reliable when the correlation is strong.

Extrapolation — estimating *outside* the range of the data; unreliable because the pattern may not continue (typing speed cannot keep rising forever).

Worked Example: For days between 4 °C and 16 °C, a line of best fit for daily heating cost y pence against temperature x °C passes through (5, 64) and (15, 24).

$$\text{Gradient} = \frac{24 - 64}{15 - 5} = \frac{-40}{10} = -4, \text{ so } y = -4x + c. \text{ Using } (5, 64): 64 = -20 + c, \text{ so } c = 84 \text{ and } y = -4x + 84.$$

At 9 °C (interpolation): $y = -36 + 84 = 48p$.

At 25 °C (extrapolation): $y = -100 + 84 = -16p$ — a negative cost is impossible, which shows why extrapolating is dangerous.

Meaning of the gradient: the cost falls by about 4p for every 1 °C rise in temperature.

Outliers

An **outlier** is a point that does not follow the pattern of the others (e.g. a kiosk that sold very few hot drinks on a cold day because it closed early). Identify it, give a possible reason in context, and leave it out when drawing the line of best fit.

Exam Tip: "Describe the relationship" needs the type of correlation **and** a sentence in context, e.g. "Negative correlation: the warmer the day, the lower the heating cost." For "Is this estimate reliable?", comment on whether it is inside the data range and how close the points are to the line.

Summary

Skill	Key point
Sampling (S1)	Random: everyone has the same chance. Bigger random samples give more reliable estimates.
Describing a population (S5)	$\frac{\text{number in sample}}{\text{sample size}} \times \text{population size}$
Pie charts (S2)	$\frac{\text{frequency}}{\text{total}} \times 360^\circ$; angles show proportions, not numbers
Pictograms (S2)	Always give a key; parts of a symbol show fractions.
Time series (S2)	Describe trend and seasonal variation; compare the same season.

Scatter graphs (S6)	Positive / negative / none; strong or weak. Correlation is not causation.
Line of best fit (S6)	Interpolation is reasonable; extrapolation is risky.